

An illustration of a woman with short brown hair, wearing a white lab coat over a green shirt and dark tie, standing in a bedroom. She is looking down at a microscope on a table. The room has a bed with a red and yellow patterned blanket, a nightstand with a lamp, and a desk with books. A blue neuron-like shape frames the scene.

Rita's first experiments were all done in her bedroom, while she and her family were in hiding from German forces during World War II.

A powerful protein

After lots of experiments, one thing became clear: the chicken embryo tissues contained some kind of instructions that told new nerve cells to grow. This was different from the theories of other scientists at the time, including Rita's former teacher!

After World War II was over, Rita was invited to repeat her experiments in a professional laboratory, where her exceptional discovery was confirmed. Rita worked with a biochemist named Stanley Cohen to find out more. They discovered a protein (an organic building block) that controls the growth and development of new nerve cells. Rita and Stanley won a Nobel Prize for discovering this "nerve growth factor." Their work has helped scientists and doctors to better understand how cells and organs grow, and what causes certain diseases, such as cancers.

A photograph of a large, ornate brick building with two prominent towers and a central archway, identified as Washington University in St. Louis. The building is surrounded by greenery and flags are visible on the towers.

Rita and Stanley worked at Washington University in St. Louis.